COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	9/8/26	**DATE SUBMITTED:**	9/10/26
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ACTIVITY NO. 7.1 **Sorting Techniques**

1. PROCEDURE OUTPUT

Code + Console Screenshot

```
const int maxSize = 100;

void displayFunc(int arr[]);

int main() {
    int data[maxSize];
    srand( Seed: time(0));

    for (int i = 0; i < maxSize; i++) {
        data[i] = rand()%5 + 1;
    }
    displayFunc(data);
```

```
void displayFunc(int arr[]) {
    for (int i = 0; i < maxSize; i++) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
}
```

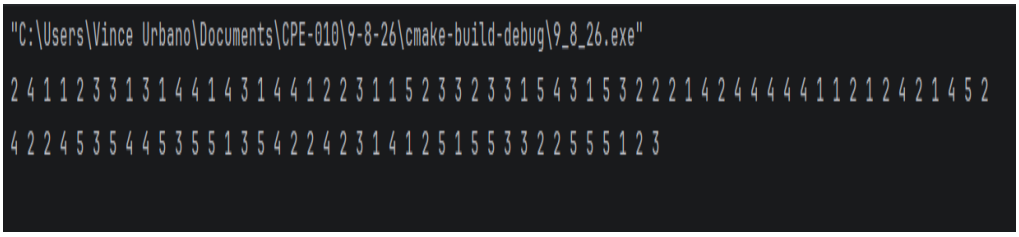
	
Observation	I have observed that it creates random 100 values from 1 to 5 using the function of rand(). It was possible to make the values from 1 to 5 because of the %5 + 1 added to the function of rand().

Table 7.1 - Array of Values for Data Sorting

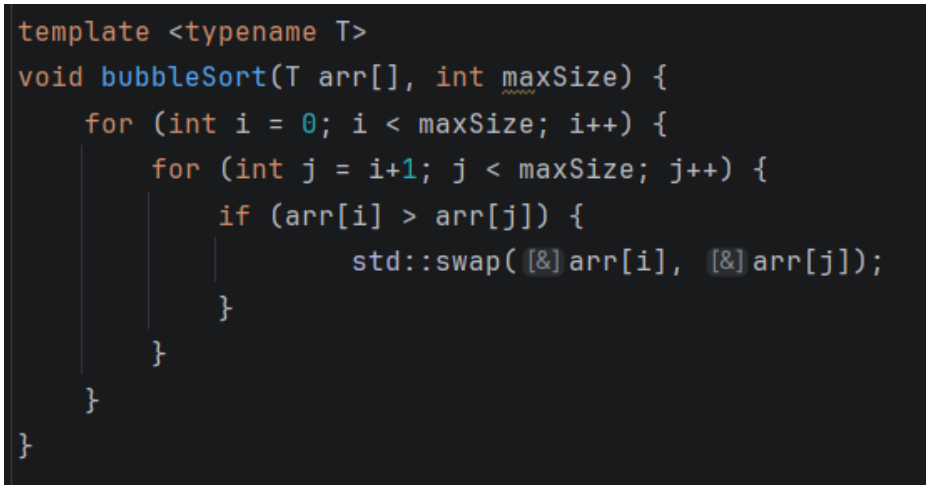
Code + Console Screenshot	
Observation	It checks the values one by one and swaps the values from the array. It can be arranged in ascending or descending order.

Table 7-2. Bubble Sort Technique

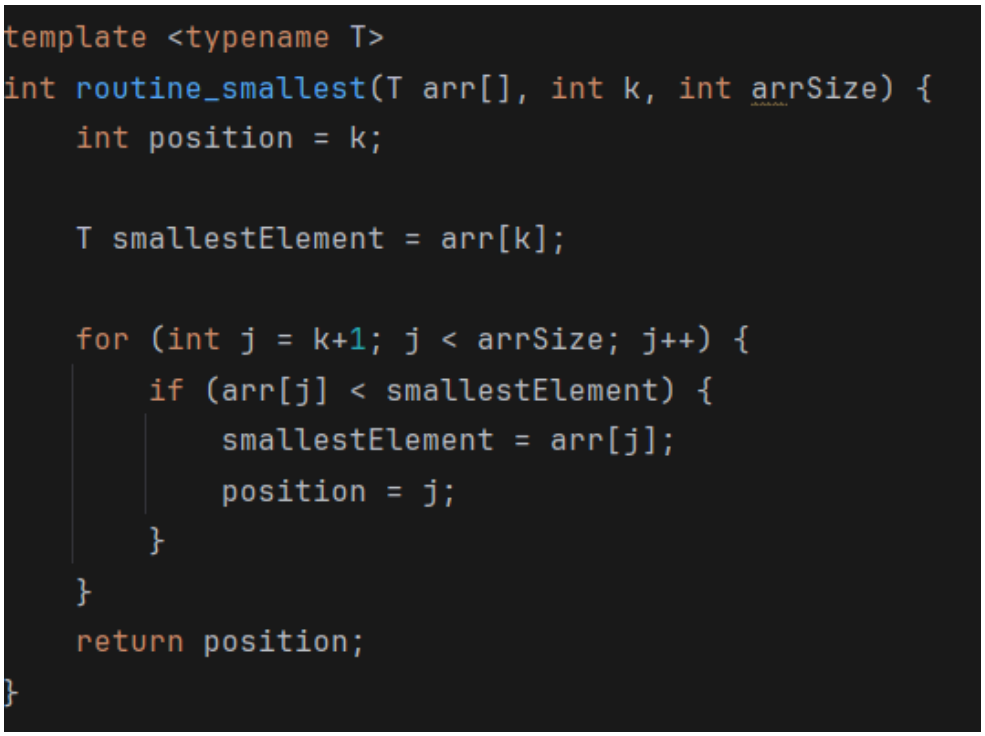
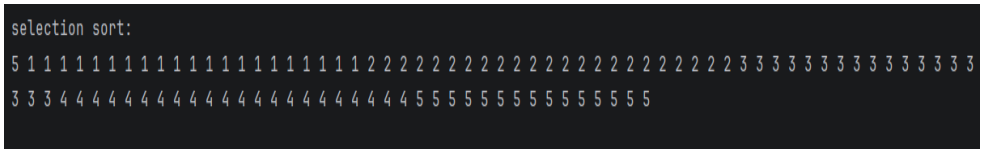
Code + Console Screenshot	  
Observation	The program is finding the lowest value by comparing two values together in this case, j and k. The smallest value will be moved to the front where the position is located.

Table 7-3. Selection Sort Algorithm

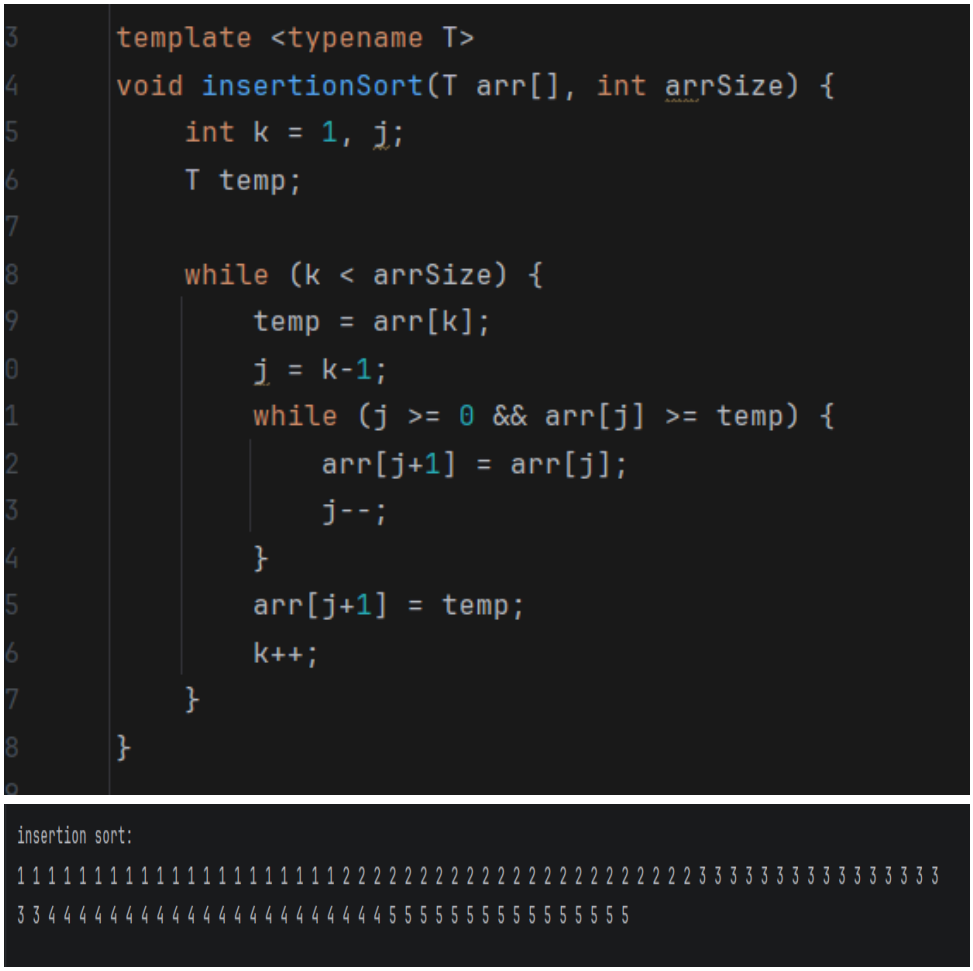
Code + Console Screenshot	
Observation	Insertion sort divides the array and swaps the position of the lowest and the largest from the array.

Table 7-4. Insertion Sort Algorithm

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Pseudocode:
Create array size
Create an array
Add random numbers from 1 -5
Sort the array
Count the number of votes made
Print the sum of each votes

Output Console Showing Sorted Array:

```
2 2 2 4 1 2 4 1 2 3 3 1 3 2 5 2 4 2 2 2 3 2 5 4 4 3 4 1 5 3 3 1 1 3 4 4 2 4 3 4 5 1 1 3 1 1 3 5 4 3 5 3 1 2 2 3 2 5 3 2
1 5 3 3 4 3 1 4 5 1 5 5 3 5 2 2 3 4 2 1 3 4 3 4 5 3 3 3 1 5 3 1 1 5 3 4 2 1 4 3

Sorted Arrays:
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5 5
```

Manual Count:
Candidate 1: 19
Candidate 2: 20
Candidate 3: 28
Candidate 4: 18
Candidate 5: 15

Count Result of Algorithm:

```
Vote Count
Number of votes for Candidate 1: 19
Number of votes for Candidate 2: 20
Number of votes for Candidate 3: 28
Number of votes for Candidate 4: 18
Number of votes for Candidate 5: 15
```

Code:

```cpp

#include <iostream>
#include "supplementary.h"
#include <time.h>
#include <cstdlib>

int maxSize = 100;
void displayFunc(int arr[]);


int main(){
    int votes[maxSize];
    srand( Seed: time(0));


    for (int i = 0; i < maxSize; i++) {
        votes[i] = rand()%5 + 1;
    }

    displayFunc(votes);
    std::cout << std::endl;
    std::cout << "Sorted Arrays: " <<std::endl;
    bubbleSort(votes, maxSize);
    displayFunc(votes);

    std::cout << std::endl;
    std::cout<< "Vote Count" <<std::endl;
```

```
}
void displayFunc(int arr[]) {
    for (int i = 0; i < maxSize; i++) {
        std::cout << arr[i] << " ";
    }
    std::cout << std::endl;
}
```

```
#ifndef supplementary_h
#define supplementary_h

template <typename T>
void bubbleSort(T arr[], int arrSize) {
    for (int i = 0; i < arrSize; i++) {
        for (int j = i+1; j < arrSize; j++) {
            if (arr[i] > arr[j]) {
                std::swap( [&] arr[i],  [&] arr[j]);
            }
        }
    }
}

template <typename T>
void voteCount(T arr[], int arrSize) {
    int count = 0;
    int cand1 = 0;
    int cand2 = 0;
    int cand3 = 0;
    int cand4 = 0;
    int cand5 = 0;

    for (int i = 0; i < arrSize; i++) {
        if (arr[i] == 1) {
            cand1++;
```

```
28            } else if (arr[i] == 2) {
29                cand2++;
30            } else if (arr[i] == 3) {
31                cand3++;
32            }else if (arr[i] == 4) {
33                cand4++;
34            }else{
35                cand5++;
36            }
37        }
38        std::cout << "Number of votes for Candidate 1: " << cand1 << std::endl;
39        std::cout << "Number of votes for Candidate 2: " << cand2 << std::endl;
40        std::cout << "Number of votes for Candidate 3: " << cand3 << std::endl;
41        std::cout << "Number of votes for Candidate 4: " << cand4 << std::endl;
42        std::cout << "Number of votes for Candidate 5: " << cand5 << std::endl;
43    }
44
45
```

Output:

```
"C:\Users\Vince Urbano\Documents\CPE-010\Sorting Supplementary\cmake-build-debug\Sorting_Supplementary.exe"
2 2 2 4 1 2 4 1 2 3 3 1 3 2 5 2 4 2 2 2 3 2 5 4 4 3 4 1 5 3 3 1 1 3 4 4 2 4 3 4 5 1 1 3 1 1 3 5 4 3 5 3 1 2 2 3 2 5 3 2
1 5 3 3 4 3 1 4 5 1 5 5 3 5 2 2 3 4 2 1 3 4 3 4 5 3 3 3 1 5 3 1 1 5 3 4 2 1 4 3

Sorted Arrays:
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5 5 5 5 5

Vote Count
Number of votes for Candidate 1: 19
Number of votes for Candidate 2: 20
Number of votes for Candidate 3: 28
Number of votes for Candidate 4: 18
Number of votes for Candidate 5: 15

Process finished with exit code 0
```

Was your developed vote counting algorithm effective? Why or why not?
I could say that my vote counting algorithm was very effective. First it sorts the array that was randomly created by the program. After it was sorted it was able to count the amount of votes per candidate.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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